



NeXtTimber[®]
by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

Acoustic Performance Guide

Version 1.0
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The image shows a modern interior space. The ceiling is made of light-colored wood planks. The walls are a neutral, light gray. A large, black-framed window or glass door is on the right, offering a view of a green lawn, a wooden fence, and trees. A yellow armchair with dark wood legs is positioned in the lower right corner. On the left wall, there is a small, square framed picture of a tree. A large green graphic with diagonal stripes is in the bottom left corner, containing the text 'Acoustic Performance Guide'.

Acoustic Performance Guide

Introduction

Timberlink Australia Pty Ltd (Timberlink) commissioned CSIRO to undertake laboratory testing on a number of systems incorporating NeXTimber® Cross Laminated Timber (CLT) that have then been assessed by PKA Acoustic Consulting (PKA). PKA have prepared an acoustic assessment and provided the predicted performance for various NeXTimber CLT wall and floor systems based on the tested results and as set out in this document. The relevant sound insulation requirements are contained in the National Construction Code (NCC), Building Code of Australia (BCA) Volume 1, Section F7 airborne sound insulation requirements.

NeXTimber CLT is a solid engineered wood product. It is well suited to multi residential apartment construction, hotels, offices and other commercial applications and is a significantly lighter construction material than the likes of concrete or steel. Structurally speaking, building lighter offers many benefits to developers in terms enabling additional levels on existing buildings, minimising strengthening works, and reduction in concrete use through thinner transfer structures and smaller foundations.¹

Building using CLT presents some acoustic challenges for developers as CLT provides less airborne and impact sound insulation compared to denser and heavier materials. To satisfy the required level of acoustic performance as outlined in the NCC, the predicted performance of NeXTimber CLT and acoustics systems proposed within this document have been achieved by adopting the following approaches:

- Adding layers of mass with the addition of board materials such as plasterboard, fibre cement and chipboard.
- Providing resilient separation between layers.
- Isolation / minimisation of rigid connection between layers.

Note that the systems proposed within this document are only relevant to NeXTimber CLT products and are intended to aid in the selection and specification of suitable acoustic systems used within building structures. The specification of the appropriate acoustic systems and appropriate approach for a project remains the responsibility of the client (including their engineers, architects and builder). The test reports used to develop the predictions within this report can be supplied if requested to aid in evaluation of systems using NeXTimber CLT.

NeXTimber Technical Information Currency

This guide is current as at the date it bears. It is the users responsibility to ensure that the latest version of this guide is in use at all times. Please check the NeXTimber website for the latest version of this guide and other NeXTimber documentation, including specific guide and test information.

Access the latest technical information at www.nexttimber.com.au

Acknowledgement of Contribution

NeXTimber would like to acknowledge the following businesses who contributed to the contents of this document:

- CSIRO Acoustic Labs
- PKA Acoustic Consulting
- REGUPOL Australia Pty. Ltd.
- Sustainable Access Floor Solutions
- Timber Development Association (TDA)

¹ Woodsolutions Guide 16 – Massive Timber Construction Systems, FWPA, Revised October 2018

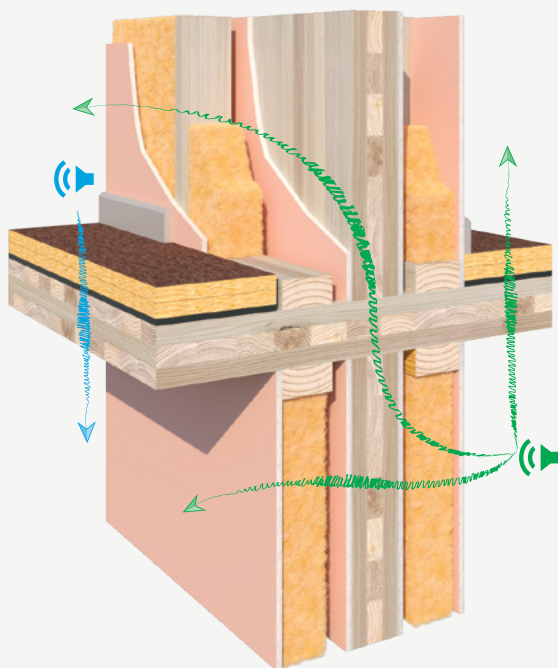
Assessment Information

This acoustic assessment is based on PKA's experience calculating the acoustic properties of lightweight wall and floor systems as well as acoustic laboratory test data as follows:

Table 1. Test References

Facility	Report Reference	Report Contents
CSIRO Acoustic Laboratory, Clayton Victoria	Airborne: TL720, tests 01 to 11 Impact: INR285, tests 01 to 11	NeXTimber CLT wall and floor systems

Figure 3. Typical Sound Transmission Paths



The acoustic laboratory testing standards are as follows:

Table 2. Acoustic Laboratory Testing Standards

Type	Standard	Description
Airborne testing	AS 1191-2002 (R2016)	Acoustics – Method for laboratory measurement of airborne sound transmission insulation of building elements
Airborne rating	AS/NZS ISO 717.1:2004	Acoustics – Rating of sound insulation in buildings and of building elements Part 1: Airborne sound insulation “ R_w ”
	ASTM E413-16	Calculation for Rating Sound Insulation “STC”
Impact testing	AS ISO 140.6 (2006) ISO 10140-3 (2010)	Acoustics – Method of laboratory measurements of impact sound insulation of floors
Impact rating	AS ISO 717.2 (2004)	Acoustics – Rating of sound insulation in buildings and of building elements Part 2: Impact sound insulation “ $L_{n,w}$ ”
	ASTM E989-89	Calculation for Impact Isolation Class “IIC”

These acoustic predictions outlined in Acoustic Assessment Sections following are based on the testing undertaken by CSIRO using NeXTimber CLT and other publicly available information. The predictions result in tolerances within $R_w \pm 2$ and $L_{n,w} \pm 2$ points within the stated ranges when validated against acoustic laboratory test results and other supporting information, which have their own inherent variability.

Product For Assessment

Table 4. NeXTimber NX3 CLT Panel Specifications

Product Name:	NeXTimber NX3 CLT Panel	
Nominal Density:	500kg/m3	
Thickness Range in Assessment:	90mm to 180mm	
Type:	Load bearing cross laminated timber	

Conditions And Limitations of Assessment

This document presents several predicted acoustic systems using NeXTimber CLT and commonly available construction products assessed by PKA. These systems and their predicted performance are made available to designers to consider when evaluating suitable floor and wall assemblies for their projects. Other acoustic solutions and systems compatible with NeXTimber may be available within the market.

The acoustic products, systems and performance contained within this document are:

- Only based on tested products and certification owned by Timberlink, tested through CSIRO using NeXTimber CLT and assessed by PKA and as described within the tables contained in the document and assume materials are installed as illustrated using good workmanship and according to material manufacturer's instructions recommended for acoustic-rated construction. Any changes to the construction of the material(s) used in the product or the system(s) can invalidate this assessment;
- Not a substitute for designers obtaining their own advice and consulting with the relevant authorities and certifiers in relation to acoustics and other aspects of design compliance relating to the products or systems structural, fire or durability performance. Before adopting

systems within a project, the acoustic details and systems should be assessed and evaluated by the projects acoustic engineer to confirm that the proposed details are fit for purpose and will meet the expectations and requirements of the project and the NCC. For example, this document makes no allowance or consideration for sound flanking that may occur in field installation and this and similar issues need to be considered by the acoustic engineer when specifying acoustic details.

- Not guarantees of field acoustic performance. They provide indicative performance based on externally tested and assessed floor systems. With appropriate design, good workmanship and attention to detail, and ideal site conditions, PKA considers field $D_{n,tw}$ and $L_{n,tw}$ performance to be broadly comparable to laboratory performance.

Please check the NeXTimber website for the latest version of this guide and other NeXTimber documentation, including specific guide and test information.

Acoustic Terminology

Airborne Sound Insulation (R_w)

Airborne sound is a sound source that originates in the air such as a person talking or a loudspeaker, as opposed to impact sound which strikes a surface such as footsteps.

Airborne sound insulation is the difference in sound pressure level between the sound entering and sound leaving a building element. The higher the value the better the sound insulation.

The R_w rating, defined as a “Weighted Sound Reduction Index”, is an acoustic laboratory measurement that determines the effectiveness of a building element’s airborne sound insulation over a range of frequencies (100Hz to 3150Hz) in a single number quantity. The R_w value, expressed in dB, is corrected for room volume and reverberation time but does not take into account sound flanking paths associated with in-situ installations.

Spectrum Adaptation Term (C_{tr})

The C_{tr} term is an airborne low frequency adjustment factor that helps quantify the low frequency performance of the building element from sources such as traffic, music, televisions etc. C_{tr} is a negative value that is added to an R_w rating.

Impact Sound Insulation ($L_{n,w}$)

Impact sound is a sound source that typically originates by striking a floor surface such as footsteps or moving furniture. Impact sounds can be transmitted through walls via lifts, plantrooms, washing machines, service pipes etc. however there is no functioning criteria for assessing these structure-borne sounds except for adopting discontinuous constructions.

Impact sound insulation is the resultant sound pressure level measured in the receive space when an ISO standard tapping machine is placed in the source space on the separating floor/ceiling. The lower the value the better the sound insulation

The $L_{n,w}$ rating, defined as “Weighted Normalised Impact Sound Pressure Level”, is an acoustic laboratory measurement to determine the effectiveness of a building element’s impact sound insulation over a range of frequencies (100Hz to 3150Hz) in a single number quantity. The $L_{n,w}$ value, expressed in dB, is corrected for room volume and reverberation time but does not take into account sound flanking paths associated with in-situ installations.

Requirements For National Construction Code

The National Construction Code (NCC), previously the Building Code of Australia (BCA), in Volume 1 Section F7 "Sound Transmission and Insulation" states that walls and floors separating places of occupancy in Class 2 and 3 buildings must "safeguard occupants from illness or loss of amenity as a result of undue sound being transmitted".

The following is a general summary the BCA sound insulation requirements for indicative purposes only. It does not substitute the need to review the NCC requirement for the project or obtain professional advice.

Table 5. Walls – Deemed-to-Satisfy Provisions

Wall Description	Airborne	Impact	NCC 2022	NCC 2019
Separating sole-occupancy units (SOUs) habitable areas	$R_w + C_{tr} \geq 50$		F7D6(1)(a)	F5.5(a)(i)
Separating SOUs wet to habitable areas	$R_w + C_{tr} \geq 50$	Discontinuous Construction	F7D6(1)(a) F7D6(1)(c)(i)	F5.5(a)(i) F5.5(a)(iii)
Separating SOUs with corridor, stairway, lobby or different classification	$R_w \geq 50$		F7D6(1)(b)	F5.5(a)(ii)
Separating SOUs with plantroom or lift shaft	$R_w \geq 50$	Discontinuous Construction	F7D6(1)(b) F7D6(1)(c)(ii)	F5.5(a)(ii) F5.5(a)(iii)
Separating SOU habitable area with services from another SOU	$R_w + C_{tr} \geq 40$		F7D7(1)(a)	F5.6(a)(i)
Separating SOU wet area with services from another SOU	$R_w + C_{tr} \geq 25$		F7D7(1)(b)	F5.6(a)(ii)
Doors separating SOU with corridor, stairway, lobby	$R_w \geq 30$		F7D6(2)	F5.5(b)

Table 6. Discontinuous Construction Requirement

Wall Type	Discontinuous Construction Requirement	NCC 2022	NCC 2019
Masonry	Wall having a minimum 20mm cavity between the 2 separate leaves, with resilient wall ties if necessary	F7D4(3)(a)	F5.3(c)(i)
Other than masonry	Wall having a minimum 20mm cavity with no mechanical linkage except at the periphery	F7D4(3)(b)	F5.3(c)(ii)

Requirements For National Construction Code

Table 7. Floors – Deemed-to-Satisfy Provisions

Floor Description	Airborne	Impact	NCC 2022	NCC 2019
Separating sole-occupancy units (SOUs)	$R_w + C_{tr} \geq 50$	$L_{n,w} \leq 62$	F7D5(1)(a)	F5.4(a)(i)
Separating SOUs with plantroom, lift shaft, corridor, stairway, lobby or different classification	$R_w + C_{tr} \geq 50$	$L_{n,w} \leq 62$	F7D5(1)(b)	F5.4(a)(ii)
Separating Class 9c sole-occupancy units	$*R_w \geq 45$	**	F7D5(2)	F5.4(b)

* The NCC 2022 Part F7 Performance Requirements F7D5 has been revised to $D_nT_w + C_{tr} \geq 40$ which approximates a design criterion of $R_w \geq 50$.

** Although the NCC 2022 Part F7 Deemed-to-Satisfy Provisions do not include an impact sound insulation rating for Class 9c aged care floors, the NCC Performance Requirements in F7P3 state that residential care buildings must achieve $L_{n,T,w} \leq 62$, which is equivalent to Class 2 and 3 Buildings.

Acoustic Assessment - Residential Walls

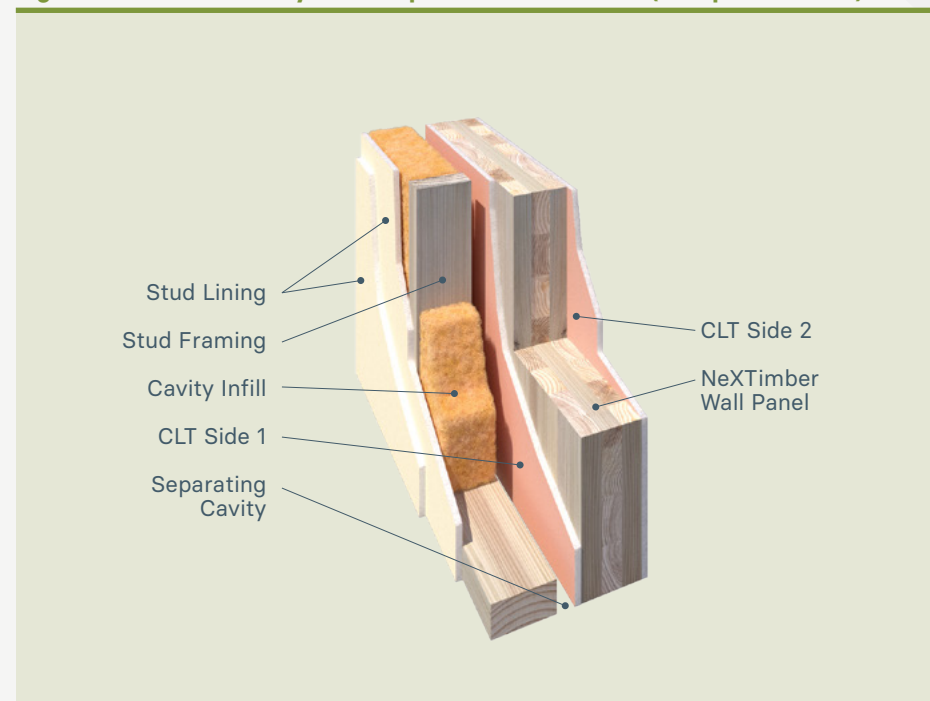
Stud Wall with Encapsulated NeXTimber CLT – Lined Both Sides

Table 8. Residential Party Wall –Separate Stud One Side (Encapsulated CLT)

Build layering	Option 1	Option 2	Option 3
Stud Lining	Min. 13mm standard plasterboard (min. 8.4kg/m ²)	Min. 2 x 13mm standard plasterboard (min. 8.4kg/m ² each)	Min. 16mm fire grade plasterboard (min. 12.4kg/m ²)
Stud Frame	Min. 70mm x 35mm timber studs at 600mm centres		
	Cavity Infill – min. 75mm glasswool insulation (min. 14kg/m ³)		
Separating Cavity	Min. 20mm gap		
CLT Side 1	Min. 16mm fire-grade plasterboard (min.12.4kg/m ²)		
NeXTimber Wall Panel	140mm (75kg/m ²)	90mm (45kg/m ²)	90mm (45kg/m ²)
CLT Side 2	Min. 16mm fire-grade plasterboard (min.12.4kg/m ²)		
Min. Wall Thickness	269mm	232mm	222mm
Predicted Acoustic Performance			
R _w	59	59	59
R _w +C _{tr}	50	51	50

Note: Refer to designer for final specification of all wall components and fixings. Stud and panel sizes may differ to satisfy strength and serviceability requirements.

Figure 9. Residential Party Wall –Separate Stud One Side (Encapsulated CLT)



Acoustic Assessment - Residential Walls

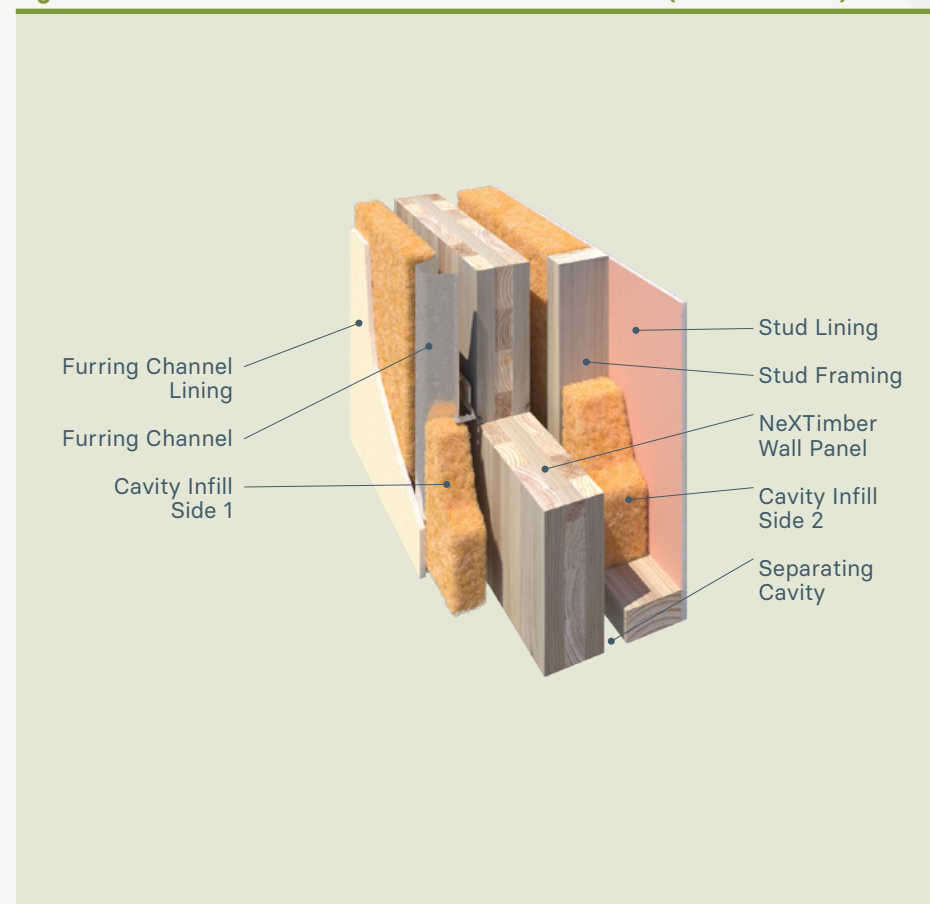
Furring Channel / Stud Wall – Lined Both Side

Table 10. Residential Service Wall – Furring Channel / Stud Wall (Concealed CLT)

Build layering	Option 1	Option 2	Option 3
Stud Lining	Min. 13mm standard plasterboard (min. 8.4kg/m ²)	Min. 2 x 13mm standard plasterboard (min. 8.4kg/m ² each)	Min. 16mm fire grade plasterboard (min. 12.4kg/m ²)
Stud Frame	Min. 70mm x 35mm timber studs at 600mm centres		
	Cavity Infill – min. 75mm glasswool insulation (min. 14kg/m ³)		
Separating Cavity	Min. 30mm gap		
CLT Side 1	Nil		
NeXTimber Wall Panel	180mm (90kg/m ²)	90mm (45kg/m ²)	90mm (45kg/m ²)
CLT Side 2	Nil		
Min. 55mm Furring Channel Cavity	Min. 28mm vertical furring channel at 600mm centres fixed with adjustable clips at 1200mm vertical centres		
	Cavity Infill – min. 50mm glasswool insulation (min. 14kg/m ³)		
Furring Channel Lining	Min. 13mm standard plasterboard (min. 8.4kg/m ²)	Min. 2 x 13mm standard plasterboard (min. 8.4kg/m ² each)	Min. 16mm fire grade plasterboard (min. 12.4kg/m ²)
Min. Wall Thickness (mm)	355	271	278
Predicted Acoustic Performance			
R _w	59	59	58
R _w +C _{tr}	50	51	51

Note: Refer to designer for final specification of all wall components and fixings. Stud and panel sizes may differ to satisfy strength and serviceability requirements.

Figure 11. Residential Service Wall – Channel and Stud Wall (Concealed CLT)



Acoustic Assessment - Residential Walls

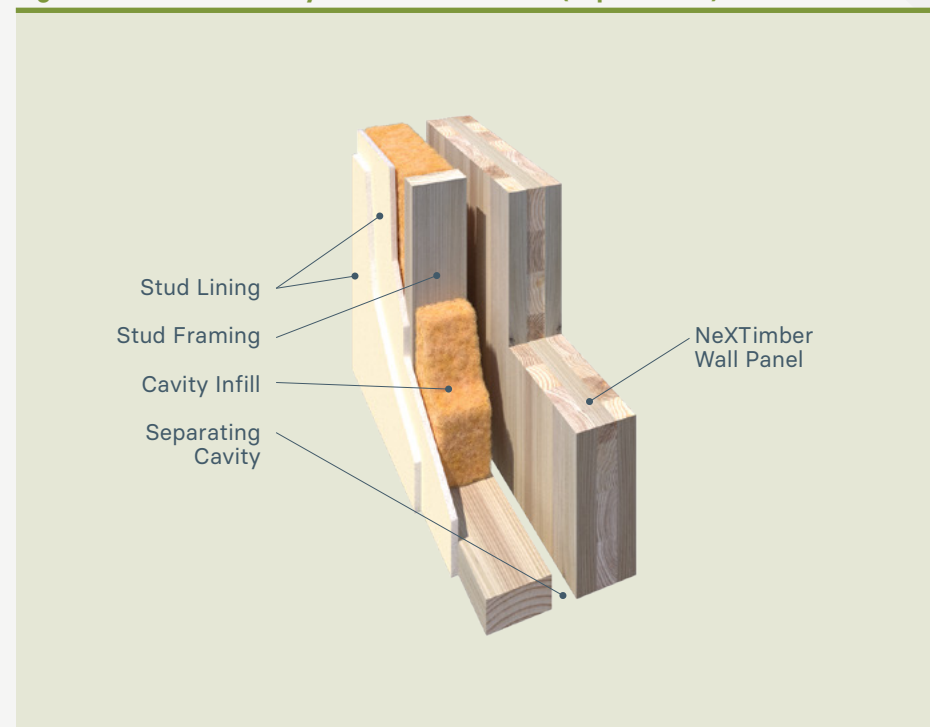
Stud Wall with Exposed NeXTimber CLT – Lined One Side

Table 12. Residential Party Wall – Stud One Side (Exposed CLT)

Build layering	Option 1	Option 2	Option 3
Stud Lining	Min. 13mm standard plasterboard (min. 8.4kg/m ²)	Min. 2 x 13mm standard plasterboard (min. 8.4kg/m ² each)	Min. 16mm fire grade plasterboard (min. 12.4kg/m ²)
Stud Frame	Min. 70mm x 35mm timber studs at 600mm centres		
	Cavity Infill – min. 75mm glasswool insulation (min. 14kg/m ³)		
Separating Cavity	Min. 20mm gap		
CLT Side 1	Nil		
NeXTimber Wall Panel	180mm (90kg/m ²)	120mm (60kg/m ²)	150mm (75kg/m ²)
CLT Side 2	Nil		
Min. Wall Thickness (mm)	307	260	280
Predicted Acoustic Performance			
R _w	59	58	59
R _w +C _{tr}	50	51	51

Note: Refer to designer for final specification of all wall components and fixings. Stud and panel sizes may differ to satisfy strength and serviceability requirements.

Figure 13. Residential Party Wall – Stud One Side (Exposed CLT)



Acoustic Assessment - Residential Walls

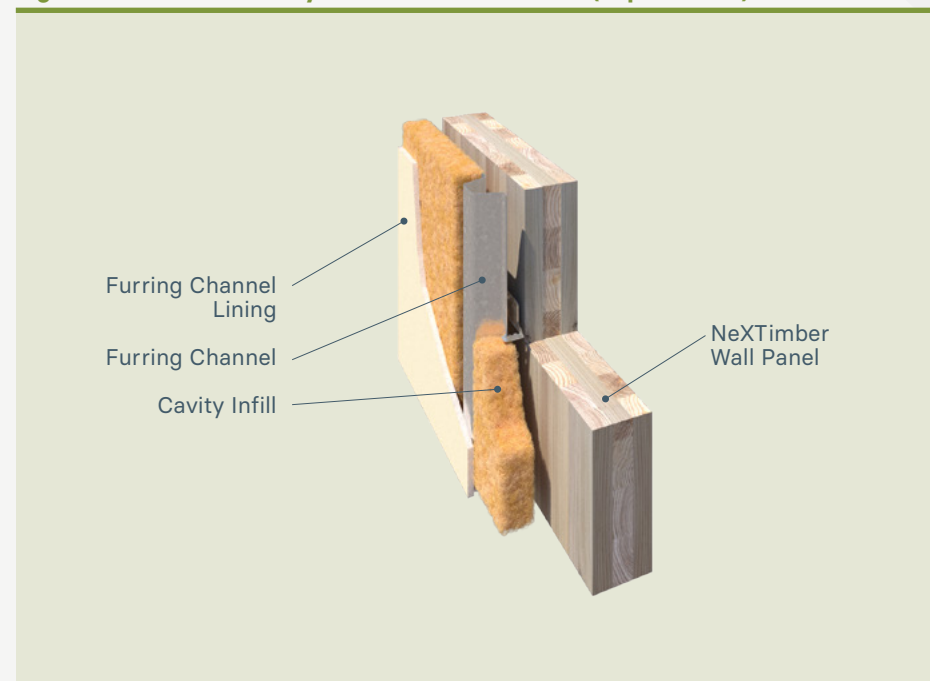
Furring Channel with Exposed NeXTimber CLT – Lined One Side

Table 14. Residential Service Wall –Channel One Side (Exposed CLT)

Build layering	Option 1	Option 2	Option 3
Furring Channel Lining	Min. 13mm standard plasterboard (min. 8.4kg/m ²)	Min. 2 x 13mm standard plasterboard (min. 8.4kg/m ² each)	Min. 16mm fire grade plasterboard (min. 12.4kg/m ²)
Min. 45mm Cavity	Min. 28mm vertical furring channel at 600mm centres fixed with adjustable resilient mounted clips at 1200mm vertical centres		
	Cavity Infill – min. 50mm glasswool insulation (min. 14kg/m ³)		
CLT Side 1	Nil		
NeXTimber Wall Panel	120mm (60kg/m ²)	90mm (45kg/m ²)	90mm (45kg/m ²)
CLT Side 2	Nil		
Min. Wall Thickness (mm)	178	161	151
Predicted Acoustic Performance			
R _w	50	53	51
R _w +C _{tr}	40	43	41

Note: Refer to designer for final specification of all wall components and fixings. Stud and panel sizes may differ to satisfy strength and serviceability requirements.

Figure 15. Residential Party Wall –Channel One Side (Exposed CLT)



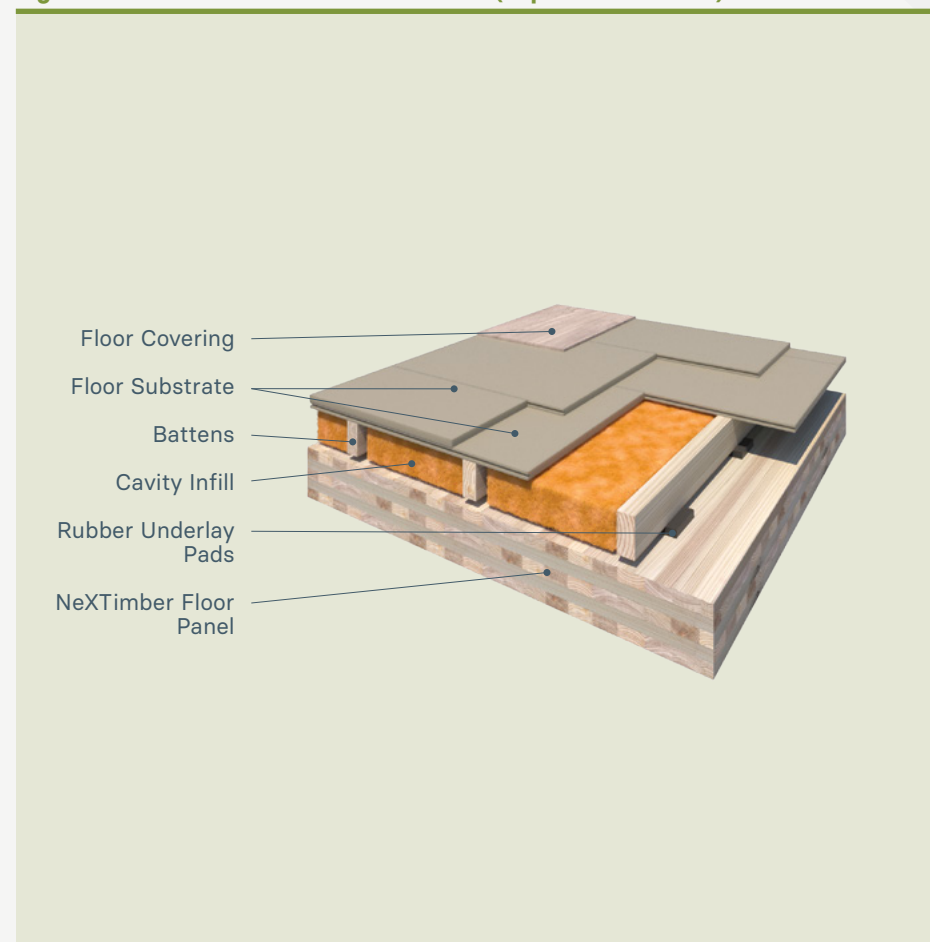
Acoustic Assessment - Residential Floors

NeXTimber CLT Floor with Exposed CLT Soffits

Table 16. Residential Floor – Raised batten (Exposed CLT Soffit)

Build layering	Option 1 Bare Floor	Option 2 Timber	Option 3 Tiles	Option 4 Carpet
Floor Covering	None	Min. 8mm laminate timber floor on min. 2mm foam underlay	Min. 10mm ceramic tiles on adhesive bed	Min. 10mm quality carpet on min. 8mm chipfoam underlay
Floor Substrate with surface density of min. 48kg/m ²	Min. 30mm sand-cement screed on support sheeting OR 2 x 19mm James Hardie Sycon Secura fibre cement (24.5kg/m ² each) OR 2 x 15mm CSR Cemintel Compressed Sheet (29.2kg/m ² each)			
Min. 100mm cavity	Min. 90mm x 45mm timber battens at 600mm min. centres with Min. 15mm thick rubber underlay pads (100mm x 100mm) at 900mm centres under battens			
Cavity Infill	Min. 90mm glasswool insulation (min. 24 kg/m ³) between battens			
NeXTimber Floor Panel	150mm (75kg/m ²)			
Ceiling side	Nil			
Min. Floor Thickness (mm)	280	290	298	298
Predicted Acoustic Performance				
R _w / R _w +C _{tr}	55/50	55/50	56/51	55/50
L _{n,w}	58	56	58	<40

Figure 17. Residential Floor – Raised batten (Exposed CLT Soffit)



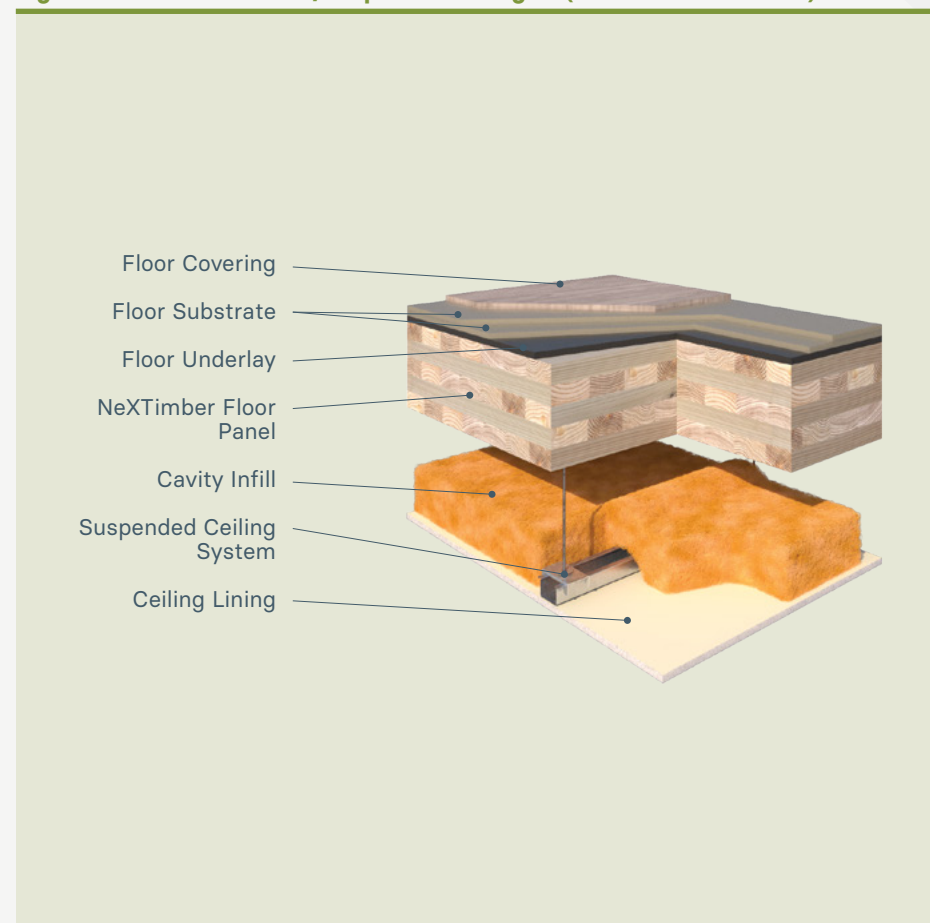
Acoustic Assessment - Residential Floors

NeXTimber CLT Floor with Suspended Ceiling

Table 18. Residential Floor/Suspended Ceilings – (Concealed CLT Soffit)

Build layering	Option 1 Bare Floor	Option 2 Timber	Option 3 Tiles	Option 4 Carpet
Floor Covering	None	Min. 8mm laminate timber floor on min. 2mm foam underlay	Min. 10mm ceramic tiles on adhesive bed	Min. 10mm quality carpet on min. 8mm chipfoam underlay
Floor Substrate with surface density of min. 26kg/m ²	2 x 9mm fibre cement sheeting (13kg/m ² each) OR Min. 20mm sand-cement screed on support sheeting			
Min. 10mm Acoustic Rubber Underlay	10mm Regupol Sonus Core 10-S (previously 6010) OR 10mm Embelton Impactamat			
NeXTimber Floor Panel	150mm (75kg/m ²)			
Min. 250mm Cavity	Suspended grid ceiling 28mm furring channels (cc 600mm)			
Cavity Infill	Min. 75mm glasswool insulation (min. 14 kg/m ³)			
Ceiling lining	13mm standard or moisture-rated plasterboard (min. 8.4kg/m ²)			
Min. Floor Thickness (mm)	441	451	459	459
Predicted Acoustic Performance				
$R_w / R_w + C_{tr}$	58/50	58/50	59/51	58/50
$L_{n,w}$	55	53	55	< 40

Figure 19. Residential Floor/Suspended Ceilings – (Concealed CLT Soffit)



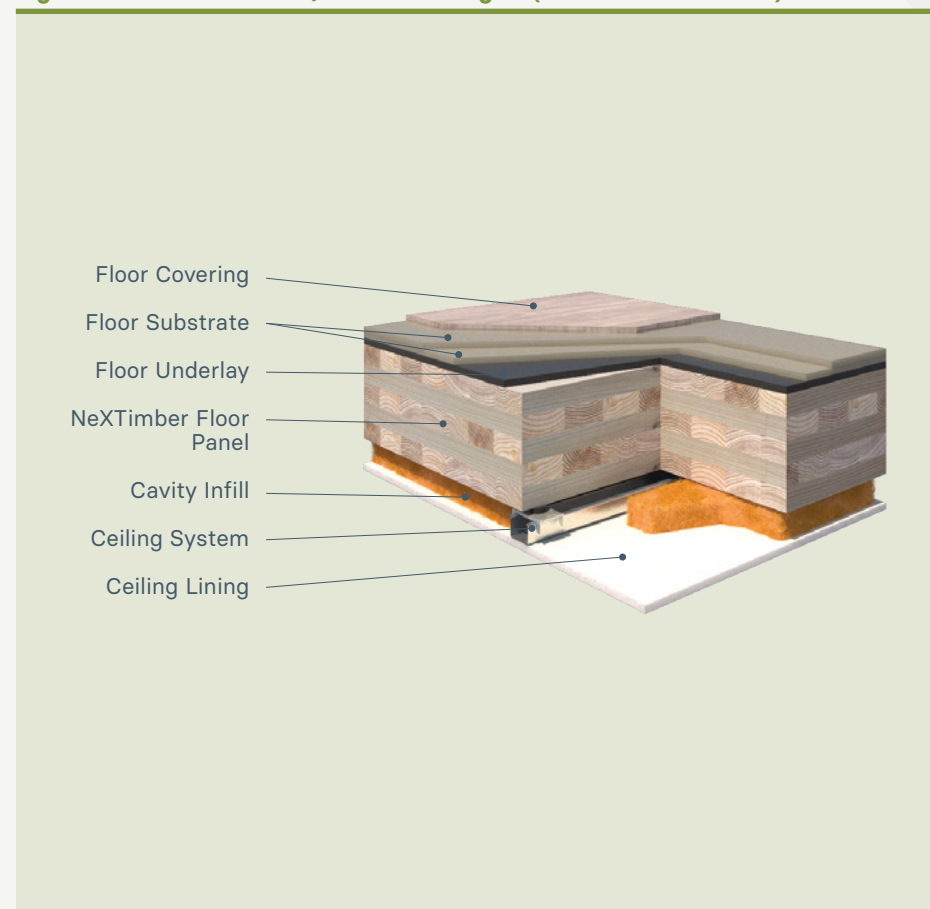
Acoustic Assessment - Residential Floors

NeXTimber CLT Floor with Resilient Mounted Ceiling

Table 20. Residential Floor/Channel Ceilings – (Concealed CLT Soffit)

Build layering	Option 1 Bare Floor	Option 2 Timber	Option 3 Tiles	Option 4 Carpet
Floor Covering	None	Min. 8mm laminate timber floor on min. 2mm foam underlay	Min. 10mm ceramic tiles on adhesive bed	Min. 10mm quality carpet on min. 8mm chipfoam underlay
Floor Substrate with surface density of min. 26kg/m ²	Min. 20mm sand-cement screed on support sheeting OR 2 x 9mm fibre cement sheeting (13kg/m ² each)			
Min. 10mm Acoustic Rubber Underlay	10mm Regupol Sonus Core 10-S (previously 6010) OR 10mm Embelton Impactamat			
NeXTimber Floor Panel	150mm (75kg/m ²)			
Min. 75mm Cavity	Resilient mounts on furring channels at 1200mm centres 28mm furring channels at 600mm centres			
Cavity Infill	Min. 50mm glasswool insulation (min.14kg/m ³)			
Ceiling lining	Min. 16mm Fire-grade plasterboard			
Min. Floor Thickness (mm)	269	279	287	287
Predicted Acoustic Performance				
$R_w / R_w + C_{tr}$	58/51	58/51	59/52	58/51
$L_{n,w}$	57	55	57	< 45

Figure 21. Residential Floor/Channel Ceilings – (Concealed CLT Soffit)



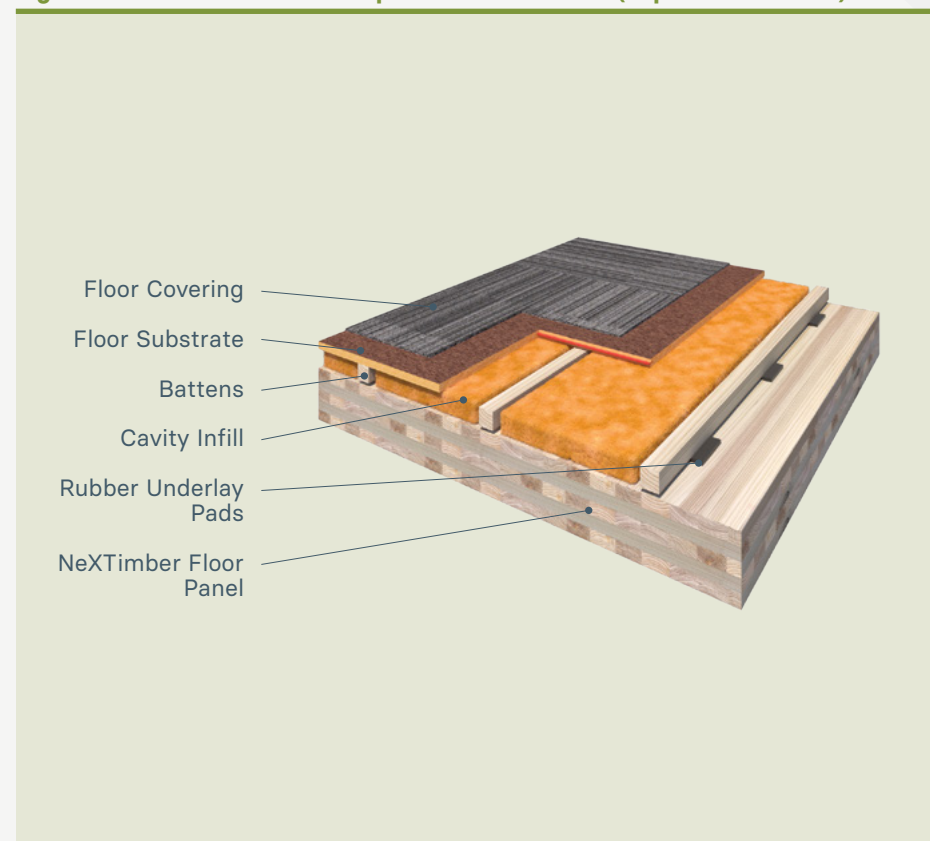
Acoustic Assessment - Commercial

Timber Batten System with Exposed NeXTimber CLT Soffit

Table 22. Commercial Floor –Raised batten (Exposed CLT Soffit)

Build layering	Option 1 Bare Floor	Option 2 Vinyl	Option 3 Carpet Tiles
Floor Covering	None	Min. 3mm vinyl	Min. 8mm carpet tiles including cushion backing
Floor Substrate with surface density of min. 14kg/m ²	22mm particleboard such as Red Tongue OR 12mm fibre cement sheet		
Min. 50mm cavity	Min. 45mm x 45mm timber battens at 450mm min. centres with Min. 10mm thick rubber underlay pads (100mm x 100mm) at 450mm centres under battens		
Cavity Infill	Min. 50mm glasswool insulation (min. 24 kg/m ³) between battens		
NeXTimber Floor Panel	150mm (75kg/m ²)		
Ceiling side	Nil		
Min. Floor Thickness (mm)	212	215	220
Predicted Acoustic Performance			
$R_w / R_w + C_{tr}$	52/44	52/44	52/44
$L_{n,w}$	64	63	55

Figure 23. Commercial Floor –Separate Stud One Side (Exposed CLT Soffit)



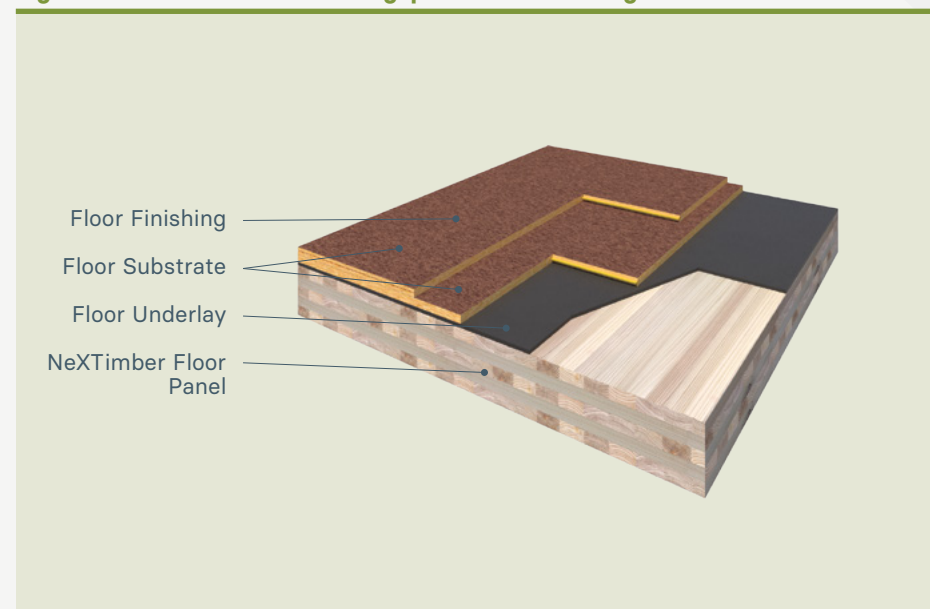
Acoustic Assessment - Commercial Floor

Rubber Underlay System with Exposed NeXTimber CLT Soffit

Table 24. Commercial Floor – Regupol with Yellow Tongue Substrate

Build layering	Option 1
Floor Covering	Bare
Floor Substrate	2 x 18mm Yellow Tongue
Acoustic Rubber Underlay	10mm Regupol
NeXTimber Floor Panel	150mm (75kg/m2)
Ceiling lining	Nil
Min. Floor Thickness (mm)	198
Predicted Acoustic Performance	
$R_w / R_w + C_{tr}$	47/40
$L_{n,w}$	67

Figure 25. Commercial Floor – Regupol with Yellow Tongue Substrate



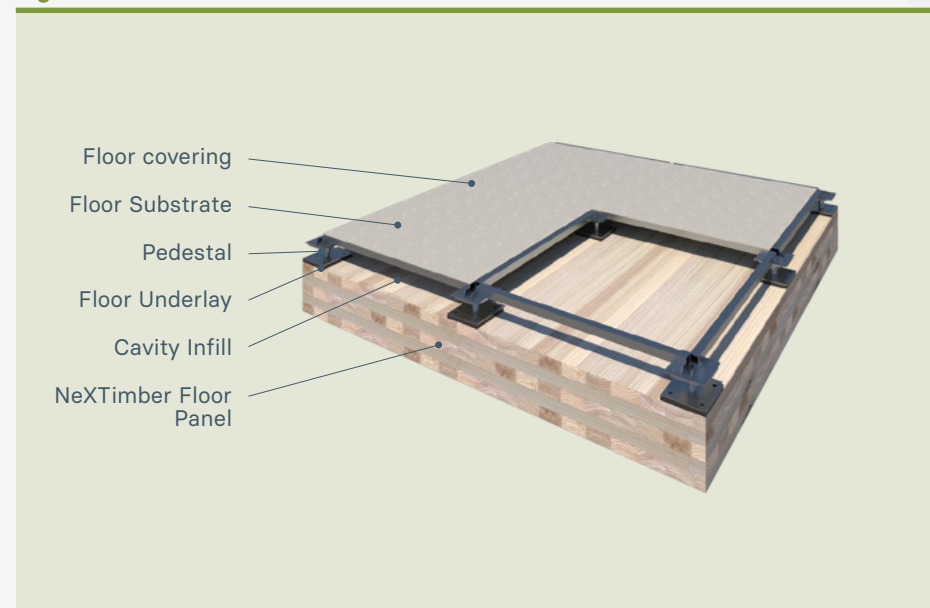
Acoustic Assessment - Commercial Floor

Raised Access Floor with Exposed NeXTimber CLT Floor Soffit

Table 26. Commercial Floor – Raised Access

Build layering	Option 1
Floor Covering	None
Raised Access Floor	600 x 600mm Tate ConCor 1500 Cementitious Tiles (44kg/m ²)
Floor Support Pedestal	600 x 600mm grid with bolted stringer
Min. 10mm Acoustic Rubber Underlay	10mm Regupol pads to each pedestal
Cavity infill	Nil
NeXTimber Floor Panel	150mm (75kg/m ²)
Ceiling lining	Nil
Min. Floor Thickness (mm)	N/A
Predicted Acoustic Performance	
$R_w / R_w + C_{tr}$	50/43
$L_{n,w}$	65

Figure 27. Commercial Floor – Raised Access



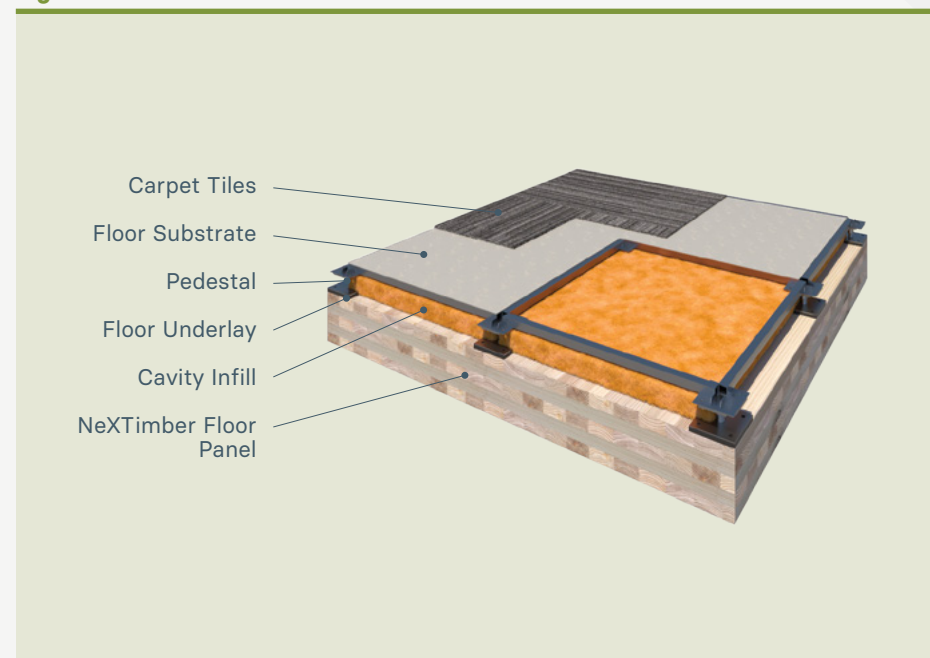
Acoustic Assessment - Commercial

Raised Access Floor with Insulation and Exposed NeXTimber CLT Floor Soffit

Table 28. Commercial Floor – Raised Access with Insulation

Build layering	Option 1	Option 2	Option 3
Floor Covering	Bare	Carpet Tiles	Vinyl
Raised Access Floor	600 x 600mm Tate ConCor 1500 Cementitious Tiles (44kg/m ²)		
Floor Support Pedestal	600 x 600mm grid with bolted stringer		
Min. 10mm Acoustic Rubber Underlay	10mm Regupol pads to each pedestal		
Cavity infill	50mm Insulation batts (24kg/m ³)		
NeXTimber Floor Panel	150mm (75kg/m ²)		
Ceiling lining	Nil	Nil	Nil
Min. Floor Thickness (mm)	243	246	251
Predicted Acoustic Performance			
$R_w / R_w + C_{tr}$	54/47	54/47	54/47
$L_{n,w}$	64	63	54

Figure 29. Commercial Floor – Raised Access with Insulation



It starts with a seed from nature, with soil,
water and sunshine added.

It's what a better tomorrow will be built on,
built out of, built with.

It's what the craftsmen, the draftsmen,
the architect, the engineer – all those who
shape tomorrow – should lean on.

If it can be renewable and store carbon, it
should, and it can.

And if it can all be done quicker, quieter,
smarter, and safer it should.

And it can.

You see, if we treat our buildings as living
breathing things, then the living breathing
things that inhabit them, will have a future
to look forward to.

Everything we do today matters, for
tomorrow will be where we all reside.

However you use NeXTimber, you won't just
be building a building.

You'll be creating a better future.

NeXTimber®
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